Numerical Verification and Proof-Guided Correction of a False Conjecture

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Abstract

Numerical search has become a routine instrument of mathematical research: before a conjecture is proved, it is tested against thousands of candidate counterexamples, and a conjecture that survives is often written down as a result. We report a case in which this practice failed, and in which the failure was found only because a human proof attempt and a machine search corrected each other. In a theory of what it costs a decision-maker to learn how its own actions change the data it later learns from, a lower bound says this cost cannot fall below a fixed floor, and we asked whether an adversary who knows the functional form of the response could beat it. An adversarial search over thousands of experimental designs said no, with a best ratio of 1.005, and we recorded the floor as structure-proof. The proof we then wrote established the bound only within two curvature lengths of the operating point; the obstruction was a single inequality that fails outside that region. Searching exactly where the proof stopped produced a counterexample: a four-point design with one far probe of weight 3×10^{-4}, attaining ratio 0.8517, verified at 50-digit precision. The original verdict was a search artifact; the corrected result is two-sided and exact. We describe the loop that made the correction visible and place the episode among eleven such corrections recorded in the same project.

1 Introduction

Machines now participate across the mathematical research loop: they suggest conjectures [Davies et al., 2021], search for counterexamples [Wagner, 2021, Romera-Paredes et al., 2024], and prove theorems [Trinh et al., 2024, Hubert et al., 2025]. Most reported successes run in one direction: the machine proposes or refutes, and a person proves. The failure direction, in which a numerical verdict is wrong and the record of how it went wrong is preserved, is rarely documented, though such records are what calibrate trust in the next verdict.

This paper documents one such failure completely, inside a body of theory about the cost of self-knowledge in performative prediction (Section 2). A machine search supported a claim. A human proof attempt could establish the claim only on a region of the design space, and it named the boundary of that region. A second machine search, restricted to the outside of that region, refuted the claim. Neither half of the loop found the result alone. We then place it among ten more **pivots**, measurement-forced changes to claims already written down, from the same project. This work is not an autonomous mathematical agent: all proofs were written by people, and what was automated is numerical checking, adversarial search over designs, and high-precision verification of witnesses (Section 3).

Contributions. (1) A reproducible account of one proof-and-search correction cycle, from the false belief through the failed proof to the frozen counterexample (Section 4, Theorem 2). (2) Ten further pivots from the same project and an assessment of what the sample supports (Sections 4 and 5). (3) The infrastructure that made the correction detectable (a theorem register with per-result proof status, preregistered checks, and continuous integration), shipped as an anonymized repository. Related work is discussed in Appendix A.

2 Setting: the cost of self-knowledge

This section gives enough of the theory to see what the counterexample of Section 4 contradicts. The object at stake is a *lower bound*, or floor: a statement that no strategy, however clever, can do better than a stated amount. Floors are valuable exactly because they are universal, which is also what makes them fragile: a single design that beats the floor falsifies it.

Consider a decision-maker whose deployed action changes the data it will later retrain on. A dealer who quotes a wide spread sees different order flow than one who quotes a tight spread, so the flow it must learn to price is partly its own doing [Nagarajan and Ashok, 2026]. This is **performative prediction** [Perdomo et al., 2020, Hardt and Mendler-Dünner, 2023]: retraining becomes a fixed-point iteration, and it converges to a point that is not the value-maximizing one. Algorithms that close this gap [Izzo et al., 2021, Miller et al., 2021, Drusvyatskiy and Xiao, 2023] need an estimate of how the world responds to the action, and they obtain it by *exploring*, that is, by deploying actions away from the current best guess and observing what changes. Those exploratory actions are real deployments, so they forgo value; the theory prices that loss.

Notation. Let h denote the scalar action, h^* the operating point the decision-maker would choose with full knowledge, and $d_t = h_t - h^*$ the deviation deployed at round t. Let ε be the *response slope*, the quantity to be estimated; σ^2 the observation noise variance; Φ the value function; and γ_{PO} its curvature at h^*, the rate at which value is lost away from h^*. The cost of exploration is the value forgone, $C_T = \sum_{t \le T} \mathbb{E}[\Phi(h^*) - \Phi(h_t)]$. Labels such as T2 are the identifiers of the project's theorem register; formal statements are in Appendix B.

Theorem 1 (Exchange rate; T2)**.** *Along any stationary exploration policy, for the least-squares slope estimator,* $\mathrm{Var}(\hat{\varepsilon}) \times C_T = \frac{1}{2}\,\gamma_{\mathrm{PO}}\sigma^2\,(1 + o(1))$*, independent of the exploration amplitude, shape, schedule, and of the contraction strength of the loop.*

Precision and cost trade off at a fixed exchange rate: exploring more reduces the variance of the estimate and raises the cost by exactly offsetting amounts, so their product is a constant. One chooses what to buy but not how much to pay. The reason is a cancellation: the information a design yields and the value it forgoes both grow with the same quantity, the spread of the deployed actions around h^*, so their ratio is fixed. Figure 1 shows the measured product lying on the predicted constant. Further results extend the floor to every policy and every estimator (T3, T4; Appendix B), and the theory was instantiated on the corporate-bond market of Nagarajan and Ashok [2026] and on a second economy (Appendix D).

3 Methodology: the verification loop

Every claim in the project passes through the following procedure, listed in the order it occurred for the pivot of Section 4 (earlier pivots used subsets of the steps). (1) **Register the claim** with an identifier and a proof status (proved, proved for a stated class, conditional on a hypothesis, or numerically supported). (2) **Preregister a numerical check:** a closed form is compared to a simulation at a tolerance fixed before the run, and rows are reported as pass, fail, or drift. (3) **Search adversarially:** for a claim of the form "every design satisfies an inequality", a random and locally refined search over designs looks for a violation, playing the role of the adversary the claim is about. (4) **Attempt a proof:** if the proof succeeds only on a sub-region, its boundary is recorded and the search of step 3 is re-run restricted to the unreached region. (5) **Verify and record:** a candidate counterexample is frozen, recomputed with `mpmath` at 50-digit precision, and added to the suite that runs on every push; any change to a written claim is recorded, with its reason, in the code where it happened.

Steps 2, 3, the re-run in 4, and the verification in 5 are automated; every proof, every choice of what to conjecture, and the decision to search outside the proof's reach were made by people. The machine never proposed a theorem; it refuted several.

4 Results: the structure-proofness pivot

The question. Theorem 1 concerns one estimator on one policy class, and the minimax floors T3 and T4 extend it to all policies and estimators. Both leave open the following. The adversary here is a hypothetical experimenter who designs deployments with the best possible information. Suppose this adversary knows the functional form of the response, not only its slope: concretely, that it is $\tau_\theta(h) = C_0 + C_1 e^{-ch}$, an exponential with three unknown parameters, where c sets the curvature and $1/c$ is the length over which the response bends. Knowing the form is real information, so it might permit an experimental design that beats the floor.

The initial search verdict. A *design* μ is a set of probe locations with weights, the fraction of the exploration budget spent at each. Its Fisher information $M(\mu)$ measures what the probes reveal about the three parameters, and the Cramér–Rao bound $g^\top M(\mu)^{-1} g$ is the smallest variance any unbiased estimator of the slope can achieve under μ, where g is the gradient of the slope with respect to the parameters. The relevant quantity is this bound times the design's cost, normalized by the floor, $R(\mu) = \left(g^\top M(\mu)^{-1} g\right) \int (h - h^*)^2 \, d\mu$, so that the floor survives structural knowledge exactly when $R(\mu) \geq 1$ for every μ. Four thousand random designs plus 2,500 local refinements returned a minimum of $R = 1.0050$, and a scan over designs pairing two near-anchor points with one distant probe found distant probes monotonically worse, reaching $R = 40.9$ at the grid's edge (Figure 2). Two independent strategies agreed, and we recorded the claim: the floor is structure-proof, with best attained ratio 1.005. This claim was later shown to be false.

The proof attempt and its reach. The proof attempt reduces $R(\mu)$ to a one-dimensional problem. It rewrites $R(\mu)$ as a ratio: the cost of the design, $\int d^2 \, d\mu$ with $d(h) = h - h^*$, divided by the smallest weighted size $\int \phi^2 \, d\mu$ of any function ϕ built from the model's sensitivities and normalized to unit slope at the anchor (Appendix C). It therefore suffices to exhibit one such ϕ that is everywhere no larger than $|d|$; then the ratio is at least one. There is a canonical candidate, the unique such function whose value, slope, and curvature at the anchor match those of d:

$$\phi_0(h) \;=\; \frac{2}{c} - e^{-c(h-h^*)}\left(\frac{2}{c} + h - h^*\right).$$

Writing $x = h - h^*$, the required domination $|\phi_0| \leq |x|$ holds for every $x \geq -2/c$, with equality only at $x = 0$ and $x = -2/c$; this is an elementary argument (Appendix C). It is also the whole story: for $x < -2/c$ the domination is not merely unproved but false, and it fails exponentially. The proof therefore established the floor for every design supported on the *reach* $[h^* - 2/c, \infty)$, from two curvature lengths below the anchor upward, and it identified the boundary exactly.

Targeted search and the witness. The failed step is not a gap in the argument but a location. We re-ran the adversarial search twice. *Inside the reach*, a hardened search (three runs of 30,000 random designs plus 10,000 refinements) pushed the minimum from 1.0050 down to $R = 1.00054$, consistent with the proof and with an infimum of exactly 1 that is approached but never attained. *Below the reach*, the same search found $R = 0.85164$. The witness is a four-point design with support $\{1.8726, 1.8665, 1.3073, 0.0500\}$ and weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$, in the market of Nagarajan and Ashok [2026] with $c = 1.5$ and anchor $h^* = 1.8461$. Every support point except the one at 0.05 lies inside the reach; the single below-reach probe of weight 3×10^{-4} is what breaks the floor. We froze the design and recomputed its ratio at 50-digit precision with `mpmath`, obtaining $0.851650\ldots$; the check now runs in continuous integration on every push.

Why the first search missed it. First, the scan's weight grid never went below 0.05, and the violating probe carries weight 3×10^{-4}. Second, the random search did generate distant probes, but always alongside a symmetric near-anchor pair, and symmetric-pair-plus-far-probe families never break the floor: their ratio approaches 1 from above as the probe weight vanishes, so the violation requires the asymmetric cluster. The earlier verdict was therefore neither noise nor a coding error but a correct summary of a search space that excluded the answer, and nothing inside that space could have revealed the exclusion.

Theorem 2 (Structure-proofness, exact reach; T9)**.** *(i) Every estimable design supported in* $[h^* - 2/c, \infty)$ *satisfies* $R(\mu) > 1$ *strictly, and* $\inf_\mu R(\mu) = 1$*, approached but not attained. Within reach, knowing the family cannot beat the floor. (ii) The reach is exact: the four-point witness above attains*

$R = 0.8517$.

Interpretation. Structure-proofness is an ignorance-of-curvature phenomenon with an exactly known extent: a design that reaches far enough below the anchor learns the curvature parameter c cheaply, and that saving is what the floor cannot account for. The floor holds because the experimenter does not know how sharply the response bends and must pay to find out; a probe placed far below the anchor, where the exponential is large, answers that question almost for free. Two checks confirm this. With c known a priori, the floor breaks outright and keeps falling with no limit the scan reaches ($R = 0.3002$ at probe $t = 1.0$, 0.0425 at $t = 0.05$, 0.0348 and still decreasing at $t = 0.001$). And the reach is exactly where the certified agent of Appendix D deploys: its trust region is $|d_t| \leq 0.8$, inside $2/c = 1.333$, so the constraint that keeps the agent safe is the one that keeps its lower bound valid. Scope: $R(\mu)$ and the floor are defined under the local-quadratic cost model (assumption A1), and under the true market cost the same witness has ratio 1.0112; the counterexample fixes the scope of the bound, not a way to beat the market.

Further pivots. One episode is an anecdote. Of the eleven pivots, three repeat the pattern above at smaller scale. **E1:** a saturation cap derived without noise failed by a factor of four against the noisy pipeline; the gap was a missing term, since retraining on noisy observations adds a stationary excitation floor of $(\sigma/\gamma)^2/(1-m^2)$ per step, itself now verified to within 0.2 to 6.4 percent (Figure 3). **E5:** an injected misspecification ae^{-2ch} had decayed to about 10^{-3} by the operating point, so the anchored fit correctly never detected it; replacing it with a linear leak the family cannot represent produced the crossover the theory predicts (Figure 4). **E7:** a claimed threefold advantage for shaped exploration collapsed to parity once the comparison arm was funded with the same energy (the unshaped arm had been given half the design's energy per step); the corrected claim is sharper, since the identification cliff is a support-and-amplitude phenomenon, not a shape phenomenon. Each replacement was stronger than the claim it displaced.

5 Discussion and conclusion

What the sample supports. On this project, numerical falsification was cheap and frequent, and it improved claims rather than only deleting them. The sample supports no claim about proof automation, other areas of mathematics, or what happens without a person deciding where to look. Unlike work in which a search refutes a conjecture directly, here the search had already returned the wrong verdict, and it was a proof's failure rather than a larger search budget that redirected it; no further search inside the original space would have found the witness.

Infrastructure. A register lists all 26 results with proof status, so a conjecture cannot be cited as a theorem. A preregistered harness compares each measurement to a closed form at a tolerance fixed before the run; the current state is 36 rows with 0 failures and one deliberately out-of-scope cell reported as drift rather than excused. A derivation suite of 38 checks runs in continuous integration; its first runs falsified three drafted claims, one with the wrong sign. Every pivot is written into the code file where it occurred. Each piece is simple; their value is in the combination, which turns a numerical verdict into a claim with a stated evidence level. We label the sharp constant $\frac{1}{2}$ in T4 as open and our own structure-proofness verdict as a search artifact: a record containing only successes cannot calibrate trust in the next verdict. This infrastructure is independent of the mathematics and is what we recommend other groups adopt.

6 Limitations

Theorem 1 is local: at finite exploration amplitude the identity is only approximate, and the deliberately out-of-scope saturating cell drifts by $+16$ percent. The value-budget floor T4 is proved only to within a factor of 13.5 of the conjectured constant. Which below-reach designs violate the floor of Theorem 2, and by how much at most, is open; and eleven pivots from one project by one group are a sample, not a study.

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A Related work

Machines and people in mathematics. The closest precedent for the shape of our loop is the use of computation to guide intuition toward statements that people then prove [Davies et al., 2021]; there the direction is machine to human, whereas one leg of our eleventh pivot runs the other way, with the proof supplying a region rather than a candidate. A second line lets search refute conjectures directly, through reinforcement learning over constructions [Wagner, 2021] or program search guided by a language model [Romera-Paredes et al., 2024]; our search is far simpler than either, and it was thorough over a space that excluded the answer. A third line automates proof itself [Trinh et al., 2024, Hubert et al., 2025, The mathlib Community, 2020]; we do none of this, and our proofs are not formalized. Our harness sits instead in the tradition of machine-assisted proof [Appel and Haken, 1977, Hales et al., 2017] and experimental mathematics [Borwein and Bailey, 2008]: it is bookkeeping about how much a numerical verdict is worth.

The substrate. The theory builds on performative prediction [Perdomo et al., 2020, Brown et al., 2022, Jagadeesan et al., 2022, Bracale et al., 2025, Lin and Zrnic, 2024], learning-while-earning [den Boer, 2015, Keskin and Zeevi, 2014, Lai and Robbins, 1979], dual control and least-costly identification [Feldbaum, 1960, Bombois et al., 2006], van Trees bounds [Gill and Levit, 1995], classical and task-optimal design [Kiefer and Wolfowitz, 1960, Pukelsheim, 2006, Wagenmaker et al., 2021, Simchowitz and Foster, 2020], and the REFLEX market [Nagarajan and Ashok, 2026, Guéant et al., 2013, Barzykin et al., 2026].

B Assumptions and formal statements

This appendix collects the standing assumptions and the formal statements of the results the body argues from. Identifiers (L1, T2, T9, and so on) are the register identifiers used throughout the supplementary repository, so each statement can be matched to its derivation document, proof, and numerical checks. The register itself, with per-result proof status, is `mlxor-derivations/THEOREMS.md`.

B.1 Standing assumptions

A1 (local quadratic scope): Φ is C^3 and strongly concave on the operating interval, with expansions taken at h^* and third-order remainder bounded by $L_3 = \sup|\Phi'''|$. **A2** (exploration class): all policies are non-anticipating; A2-sym adds $\mathbb{E}[u_t \mid \mathcal{F}_t] = 0$, A2-adaptive is unrestricted, and costs are measured from the certainty-equivalent anchor. **A3** (noise exogeneity): the observation noise ζ_t is i.i.d., mean zero, variance σ^2, independent of $\mathcal{F}_t$. **A3′** relaxes this to a feedback channel with gain ϕ. **A4** (trust region): $|d_t| \le r$ pathwise. **A5** (stationary scale): $S_T = \Theta(T)$. **A6** (stable base loop): the retraining modulus satisfies $m < 1$.

The local dynamics are $d_{t+1} = -m\, d_t + u_t$ with $d_t = h_t - h^*$, observations $\tau_t = \tau(h^*) - \varepsilon d_t + \zeta_t$, and Φ satisfying $\Phi'(h^*) = \delta_0$ and $-\Phi'' = \gamma_{\mathrm{PO}}$ at the operating point. S_{xx} denotes the centered design energy $\sum_{t\le T}(d_t - \bar{d})^2$.

B.2 The exchange rate and its floors

Lemma 1 (Cost equivalence; L1)**.** *Under A1–A3 with conditionally symmetric exploration, the expected incremental performative cost satisfies* $C_T = \frac{1}{2}\gamma_{\mathrm{PO}}\,\mathbb{E}\big[\sum_{t\le T} d_t^2\big] + R_T$ *with* $|R_T| \le \frac{L_3}{6}\mathbb{E}\sum_t |d_t|^3$*, and the realized cost's variance decomposes exactly as* $\mathrm{Var}(C_T^{\mathrm{real}}) = \delta_0^2\,\mathrm{Var}\left(\sum_t d_t\right) + \frac{\gamma_{\mathrm{PO}}^2}{4}\,\mathrm{Var}\left(\sum_t d_t^2\right)$ *with vanishing cross term.*

The body's Theorem 1 is the asymptotic form of the following pathwise identity, which is what is proved in Appendix C.

Theorem 3 (The exchange rate, pathwise; T2)**.** *Under A1, A2-sym, A3 and A6, in the stationary regime, on every realization,*

$$\mathrm{Var}(\hat{\varepsilon} \mid \textit{design}) \times \tfrac{1}{2}\gamma_{\mathrm{PO}} \sum_{t\le T} d_t^2 = \tfrac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\Big(1 + \frac{T\bar{d}^2}{S_{xx}}\Big) = \tfrac{1}{2}\gamma_{\mathrm{PO}}\sigma^2\big(1 + O_P(1/T)\big),$$

independent of the exploration intensity, the horizon, and the modulus.

Theorem 4 (Minimax lower bound, deviation budget; T3)**.** *Over all non-anticipating exploration policies with* $\sum_{t \le T} d_t^2 \le D$ *almost surely and all estimators, biased included, the minimax risk over an interval of width* w *satisfies* $\inf \sup_\varepsilon \mathbb{E}[(\hat{\varepsilon} - \varepsilon)^2] \ge \sigma^2/(D + \sigma^2(\pi/w)^2)$*, so the exchange-rate product is a floor, attained with equality by symmetric probing with least squares. The argument is a van Trees reduction in the style of Gill and Levit [1995].*

Theorem 5 (Minimax lower bound, value budget; T4)**.** *Under stationary-scale exploration and the certainty-equivalent anchor, for every non-anticipating policy and every estimator,* $\sup \mathrm{Var}(\hat{\varepsilon}) \times C_T \ge \gamma_{\mathrm{PO}} \sigma^2/27\,(1 - O(T^{-1/2}))$*, the supremum over the two-point response family at the minimax scale, the constant arising from a Le Cam reduction optimized at* $c = 2/3$*. The sharp constant* $\frac{1}{2}$*, against which no simulated policy fell by more than Monte Carlo error, is conjectured and is part of the open problem OPEN-1. It is not proved here or anywhere in the repository.*

B.3 Structure-proofness

Theorem 6 (Structure-proofness of the floor, exact reach; T9)**.** *Fix the structural family* $\tau_\theta(h) = C_0 + C_1 e^{-ch}$ *(*$C_1 > 0$*,* $c > 0$*), the anchor* $h^* > 0$*, the sensitivity* $s(h) = \partial_\theta \tau_\theta(h)$*, and the estimand gradient* $g = \partial_\theta \varepsilon(h^*) = -s'(h^*)$*. For a finitely supported design* μ *define the normalized Cramér–Rao product*

$$R(\mu) = \left(g^\top M(\mu)^{-1} g\right) \int (h - h^*)^2 \, d\mu, \qquad M(\mu) = \int s\, s^\top \, d\mu,$$

so that $R(\mu) \ge 1$ *states that the family-knowing Cramér–Rao product* $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ *is at least the exchange-rate floor* $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$ *under the local-quadratic (A1) cost, for unbiased estimators and, in the local-asymptotic-minimax sense, for all regular estimators. (i) Within reach the floor is structure-proof: if* $\mathrm{supp}(\mu) \subset [h^* - 2/c, \infty)$*, then either* $\varepsilon(h^*)$ *is not estimable under* μ*, in which case the bound is infinite, or* $R(\mu) \ge 1$*, strictly whenever* μ *places mass off the two equality points* $\{h^*, h^* - 2/c\}$*, which is every estimable design, since a design supported on those two points alone leaves* g *outside the range of* $M(\mu)$*; and* $\inf_\mu R(\mu) = 1$*, approached but not attained by symmetric three-point designs collapsing at* h^**. In particular the floor binds on every trust-region design with* $r \le 2/c$*. (ii) The reach is exact: the pointwise domination underlying (i) fails strictly below* $h^* - 2/c$*, and an explicit four-point design with a single below-reach probe of weight* 3×10^{-4} *attains* $R = 0.8517$*, computer-verified at 50-digit precision as check V9.2.*

B.4 Other register results

The register also contains T1 (information saturation of the noiseless retraining trajectory), T5a and C5.1 (A-optimal exploration under a value budget and the price of isotropic jitter), T6 (Chebyshev unidentifiability of the curvature c from retraining trajectories), T7 (the anchoring crossover), L4 (perturbed modulus under an estimated response), and P7.1 (safety under pessimism given a confidence sequence). These are stated and proved in the supplementary repository (`mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents) and are not restated here, since the body does not argue from them; they appear in the body only as the context that Theorems 1 and 2 sit in, and in Table 1 as verification rows.

C Proofs

This appendix proves the two results the body argues from: the exchange rate of Theorem 1, whose exact pathwise form is Theorem 3, and the two-sided structure-proofness result of Theorem 2, stated in full as Theorem 6. Both proofs are reproduced from the derivation repository without change; the check identifiers V2.1 and V9.1 to V9.4 refer to its verification suite.

C.1 The cost-equivalence lemma

Proof of Lemma L1 (cost equivalence). Taylor's theorem with Lagrange remainder gives, for each t,

$$\Phi(h^*) - \Phi(h_t) = -\delta_0 d_t + \tfrac{1}{2}\gamma_{\mathrm{PO}} d_t^2 + r_t, \qquad |r_t| \le \tfrac{L_3}{6}|d_t|^3,$$

with $L_3 = \sup|\Phi'''|$ on the operating interval. Summing over $t \le T$ and taking expectations, it remains to show $\mathbb{E}[\sum_t d_t] = 0$ under A2-sym. The recursion gives $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = -m\, d_t + \mathbb{E}[u_t \mid \mathcal{F}_t] = -m\, d_t$, so $\mathbb{E}[d_{t+1}] = -m\,\mathbb{E}[d_t]$ by the tower property, and $\mathbb{E}[d_t] = (-m)^t\mathbb{E}[d_0] = 0$ when the loop starts at the operating point or in the stationary regime. The quadratic term contributes $\frac{1}{2}\gamma_{\mathrm{PO}}\mathbb{E}[\sum_t d_t^2]$, which is the claim.

For the variance decomposition, write the realized cost (net of the cubic remainder) as $C = -\delta_0 X + \frac{1}{2}\gamma_{\mathrm{PO}} Y$ with $X = \sum_t d_t$, $Y = \sum_t d_t^2$. Then $\mathrm{Var}(C) = \delta_0^2\,\mathrm{Var}(X) + \frac{\gamma_{\mathrm{PO}}^2}{4}\,\mathrm{Var}(Y) - \delta_0\gamma_{\mathrm{PO}}\,\mathrm{Cov}(X,Y)$. Under Gaussian innovations the vector $(d_1,\dots,d_T)$ is jointly Gaussian with mean zero, and every third moment of a zero-mean Gaussian vector vanishes; since $\mathrm{Cov}(X,Y) = \sum_{t,s}\mathbb{E}[d_t d_s^2]$ is a sum of third moments, it is zero. (For non-Gaussian symmetric innovations the same conclusion follows from the sign symmetry $d \mapsto -d$ of the stationary law.) Verified: V0.1–V0.3, including the necessity check that an asymmetric rule breaks the expectation identity by exactly $-\delta_0\mathbb{E}[\sum_t d_t]$. $\square$

C.2 The exchange rate

Proof of Theorem T2 (pathwise exchange rate). Under A3, conditionally on the design $\{d_t\}$ the observations are independent with means linear in ε and variance σ^2, so the least-squares slope has conditional variance $\mathrm{Var}(\hat\varepsilon \mid \text{design}) = \sigma^2/S_{xx}$ (Gauss–Markov). Multiplying by the quadratic cost $\frac{1}{2}\gamma_{\mathrm{PO}}\sum_t d_t^2$ and using $\sum_t d_t^2 = S_{xx} + T\bar d^2$ gives the displayed identity exactly, on every realization. In the stationary regime $\bar d = O_P(T^{-1/2})$ by the stationary long-run variance $\mathrm{Var}(\sum_{t\le T} d_t) = Tv(1-m)/(1+m)\,(1+o(1))$, while $S_{xx}/T \to v$ a.s., so $T\bar d^2/S_{xx} = O_P(1/T)$. Verified: V2.1 (machine precision). $\square$

C.3 Structure-proofness, and the exactness of the reach

Proof of Theorem T9 (structure-proofness within reach). Write $x = h - h^*$, $d(h) = x$, and $V = \mathrm{span}\{1, e^{-ch}, he^{-ch}\}$, the span of the components of s (a bijective correspondence $u \leftrightarrow \phi_u = u^\top s$ between $\mathbb{R}^3$ and V, since $C_1 \ne 0$). Differentiating s gives $s'(h) = (0, -ce^{-ch}, C_1e^{-ch}(ch-1))^\top$, so $g = -s'(h^*)$ componentwise, which is the stated estimand gradient.

Step 1: one-dimensional reduction. For $M(\mu) \succ 0$ the generalized Rayleigh identity gives

$$g^\top M(\mu)^{-1} g = \sup_{u\ne 0}\frac{(u^\top g)^2}{u^\top M(\mu) u} = \sup_{\phi\in V}\frac{\phi'(h^*)^2}{\int\phi^2\,d\mu} = \Big(\inf\Big\{\int\phi^2\,d\mu : \phi\in V,\ \phi'(h^*) = 1\Big\}\Big)^{-1},$$

using $u^\top M u = \int\phi_u^2\,d\mu$ and $u^\top g = -\phi_u'(h^*)$ (the sign is lost in the square). Hence

$$R(\mu) = \frac{\int d^2\,d\mu}{\inf\{\int\phi^2\,d\mu : \phi\in V,\ \phi'(h^*) = 1\}}.$$

Step 2: the candidate and its pointwise domination. Let

$$\phi_0(h) = \frac{2}{c} - e^{-cx}\Big(\frac{2}{c} + x\Big) = \frac{2}{c}\cdot 1 - e^{ch^*}\Big(\frac{2}{c} - h^*\Big)e^{-ch} - e^{ch^*}\,he^{-ch} \in V,$$

the unique element of V with 2-jet $(0,1,0)$ at h^* (the 2-jet map is invertible on V: in the basis $\{1, e^{-ch}, he^{-ch}\}$ its determinant is $c^2e^{-2ch^*} \ne 0$); the expansion is $\phi_0(x) = x - (c^2/6)x^3 + O(x^4)$. Writing $f(x) = \frac{2}{c} - e^{-cx}(\frac{2}{c} + x)$ we claim

$$|f(x)| \le |x| \text{ for all } x \ge -2/c, \text{ with equality iff } x \in \{0, -2/c\},$$

and $|f(x)| > |x|$ strictly for every $x < -2/c$. Indeed: (a) $v(x) = f(x) - x$ has $v(0) = 0$ and $v'(x) = e^{-cx}(1+cx) - 1 < 0$ for $x \ne 0$ (since $1 + t < e^t$ for $t = cx \ne 0$); $v' \le 0$ with equality only at the isolated point 0 makes v strictly decreasing on $\mathbb{R}$, so $f > x$ on $x < 0$ and $f < x$ on $x > 0$. (b) $f(0) = 0$ and $f'(x) = e^{-cx}(1+cx) > 0$ for $x \ge 0$, so $f \ge 0$ on $[0,\infty)$ and, with (a), $0 \le f \le x$ there. (c) $q(x) = f(x) + x$ has $q(-2/c) = 0 = q(0)$ and $q'(x) = e^{-cx}(1+cx) + 1$, where $r(x) = e^{-cx}(1+cx)$ is strictly increasing on $x < 0$ ($r'(x) = -c^2xe^{-cx} > 0$) from $r(-2/c) = -e^2 < -1$ to $r(0) = 1$; so q' changes sign exactly once on $(-2/c, 0)$, from negative to positive, and $q \leq 0$ there with equality only at the endpoints; with (a) ($f \geq x$, equality only at 0) this gives $x \leq f \leq -x$ on $[-2/c, 0]$. (d) For $x < -2/c$, $r(x) < -e^2$ gives $q' < 0$, so moving left from $-2/c$ the function q increases above 0: $f(x) > -x = |x| > 0$, and in fact $f(x) = 2/c + e^{c|x|}(|x| - 2/c)$ grows exponentially.

Step 3: the bound. If $\operatorname{supp}(\mu) \subset [h^* - 2/c, \infty)$, Step 2 gives $\int \phi_0^2 \, d\mu \leq \int d^2 \, d\mu$, so by Step 1, $R(\mu) \geq \int d^2 \, d\mu / \int \phi_0^2 \, d\mu \geq 1$. Strictness: equality in Step 2 holds only at $x \in \{0, -2/c\}$, so any μ placing positive mass off $\{h^*, h^* - 2/c\}$ (in particular any with three distinct support points) has $\int \phi_0^2 \, d\mu < \int d^2 \, d\mu$ and $R(\mu) > 1$. If $M(\mu)$ is singular, two cases: $g \notin \operatorname{range} M(\mu)$ makes $\varepsilon(h^*)$ non-estimable (infinite bound); $g \in \operatorname{range} M(\mu)$, which does occur for special two-point designs (a one-parameter family of in-reach pairs; their finite pseudo-inverse products were checked to obey the bound, e.g. $R = 1.063$ at the pair $\{1.627, 2.127\}$), keeps the sup-representation valid over $\{u : u^\top M u > 0\}$, and $\int \phi_0^2 \, d\mu = 0$ would put $u_0 \in \operatorname{null} M$ with $u_0^\top g = -1 \neq 0$, contradicting $g \in \operatorname{range} M$; so $\int \phi_0^2 \, d\mu > 0$ and the bound goes through verbatim. Throughout, the Cramér–Rao functional bounds unbiased estimation, and all regular estimators in the local-asymptotic-minimax sense.

Step 4: the infimum is 1. The Cramér–Rao functional is invariant under C^1 reparametrization, and the 2-jet map $\theta \mapsto (\tau_\theta(h^*), \varepsilon(h^*), \tau_\theta''(h^*))$ is a local diffeomorphism: its Jacobian $[\, s(h^*) \;\; -s'(h^*) \;\; s''(h^*) \,]$ has determinant $C_1 c^2 e^{-2ch^*} \neq 0$ (direct computation; V9.4). In jet coordinates η the model reads $\tau = \eta_1 - \eta_2 x + \frac{1}{2}\eta_3 x^2 + O(x^3)$ with sensitivity $\tilde{s}(x) = (1, -x, x^2/2)^\top + O(x^3)$, and the estimand is η_2. For the symmetric three-point design μ_δ uniform on $\{h^* - \delta, h^*, h^* + \delta\}$, scale by $D = \operatorname{diag}(1, \delta, \delta^2)$: $M_\eta(\mu_\delta) = D \hat{H}_\delta D$ with $\hat{H}_\delta \to \hat{H}_0$ invertible (the exact quadratic-model moment matrix), so $(M_\eta^{-1})_{22} = \delta^{-2}(\hat{H}_\delta^{-1})_{22}$ and $R(\mu_\delta) \to R_{\text{quad}}$. In the exact quadratic model, for any symmetric design (odd moments $m_1 = m_3 = 0$), the $(2,2)$ cofactor gives $(M_\eta^{-1})_{22} = 1/m_2$ and $R_{\text{quad}} = m_2 \cdot (1/m_2) = 1$. Hence $R(\mu_\delta) \to 1$ (measured rate $R(\mu_\delta) - 1 = O(\delta^2)$; V9.2), the infimum is 1, and by Step 3 it is not attained.

Step 5: the reach is exact. Below $h^* - 2/c$ the domination of Step 2 fails strictly, and the failure is realized by a design, not only by the candidate: the frozen four-point witness (support $\{1.8726, 1.8665, 1.3073, 0.0500\}$, weights $\{0.003329, 0.955702, 0.040697, 0.000272\}$ after normalization, in the register market $c = 1.5$, frozen anchor $h^* = 1.8461$) has $R = 0.851650\ldots$, verified at 50-digit precision (`mpmath`) and reproduced in float64 by `run_open1.py` (V9.2). All its support except the probe at 0.05 lies within reach: one vanishing-weight below-reach probe breaks the floor. Symmetric pair-plus-far-probe families do not break it (their ratio approaches 1 from above as the probe weight vanishes); the violation requires the asymmetric cluster, which is why coarse searches missed it. Verified: V9.1–V9.4. □

D Verification rows

Table 1 lists selected rows of the preregistered harness. Appendix E plots the rows the body refers to.

Table 1: Selected rows from the preregistered check harness. The full table (36 rows, 0 FAIL), all seeds, tolerances, and code are in the supplementary repository. The drift row is a deliberately out-of-scope cell, measured and reported rather than excused.

Claim (register)	Measured	Predicted	Verdict
Exchange rate, $m = 0.3$, two amplitudes (T2)	0.01062, 0.01070	0.01056	pass
Exchange rate, $m = 0.6$, two amplitudes (T2)	0.00549, 0.00569	0.00535	pass
Saturating environment (outside A1)	$+16\%$ over rate	out of scope	drift
Feedback floor, $m = 0.3/0.6/0.85$ (T1 corollary)	within 0.2–6.4%	$(\sigma/\gamma)^2/(1-m^2)$	pass
Design arm sd($\hat{c}$); jitter/design ratio (T6)	0.234; 7.8×	Fisher 0.248; ≥ 3	pass
Certified agent reaches h_{PO}; trust region (L4)	1.258; 0 violations	1.296; 0	pass
Isotropic/A-optimal risk ratio, $d = 8$ (T5a)	1.529	$F = 1.506$	pass
Nonparametric MSE at (B, T) optimum (T7)	0.0132	0.0123	pass
Second economy, 4 cells (T2, exact in LQ); agent	0.384, 0.192; 12.65	0.384, 0.192; 12.5	pass
Multi-book shaped/isotropic steady-state risk (T5a)	1.20 (flat control 1.00)	> 1.15 (≈ 1)	pass
Real-data calibration port, 10 cells	10/10 reproduce	within 0.8%	pass
Within-reach min CR ratio; witness beyond (T9)	1.0005; 0.8517	≥ 1; < 1	pass

E Figures

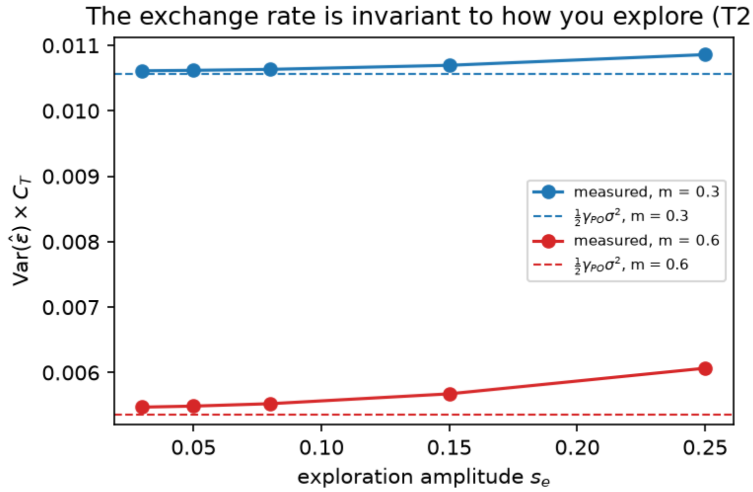

Figure 1: The exchange rate is invariant to how one explores (T2). Measured $\mathrm{Var}(\hat{\varepsilon}) \times C_T$ against exploration amplitude at two contraction moduli m; the dashed lines are the predicted constant $\frac{1}{2}\gamma_{\mathrm{PO}}\sigma^2$. The product does not depend on how much one explores, only on the curvature and the noise.

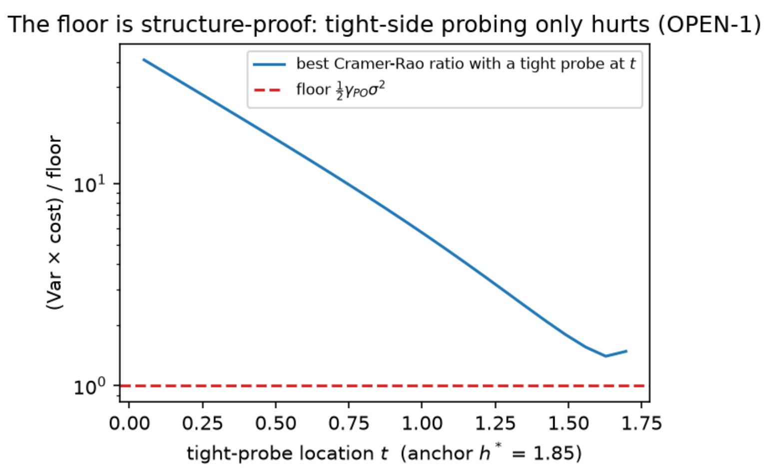

Figure 2: The search that produced the false verdict (OPEN-1). Best Cramér–Rao ratio with one tight probe at location t, over the coarse weight grid with $w \geq 0.05$; the floor is the dashed line. On this family, tight probes are monotonically worse the further they sit from the anchor, which is what we believed in general. The counterexample of Section 4 lies outside this family: its probe weight is 3×10^{-4}, inside an asymmetric cluster the search never generated.

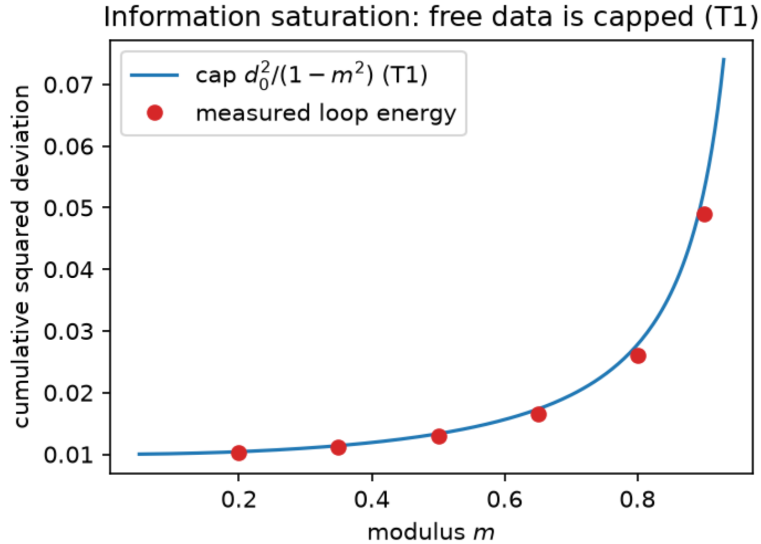

Figure 3: Information saturation (T1). Cumulative squared deviation of the noiseless retraining trajectory against the modulus m; the cap $d_0^2/(1-m^2)$ (line) matches the measured loop energy (points). Without deliberate exploration, the information a retraining loop reveals about its own performativity is bounded, and the bound grows with m. Pivot E1 added to this picture the excitation floor $(\sigma/\gamma)^2/(1-m^2)$ that noisy observations contribute (Table 1, feedback-floor row).

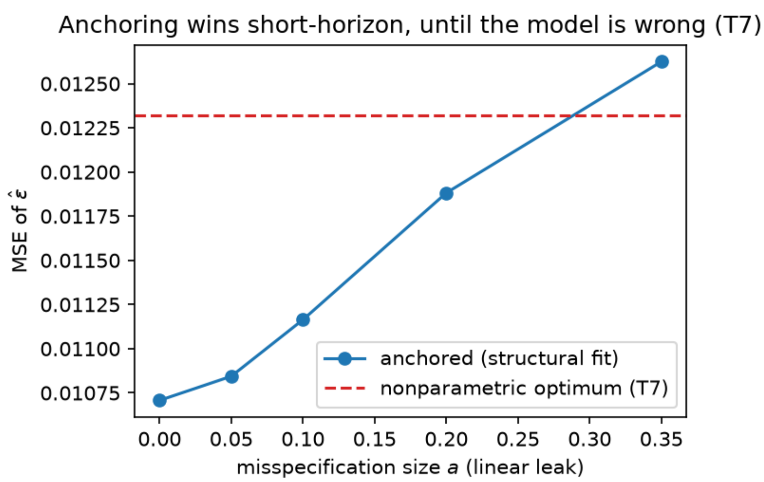

Figure 4: Anchoring crossover (T7). Mean squared error of the anchored structural fit against the size a of a linear misspecification the family cannot represent, compared with the budget-and-horizon-optimal nonparametric estimator (dashed). Anchoring wins at small misspecification and loses past the crossover near $a \approx 0.29$, as T7 predicts. Pivot E5 replaced an exponential misspecification, which had decayed to about 10^{-3} by the operating point and was correctly invisible to the fit, with this linear leak.

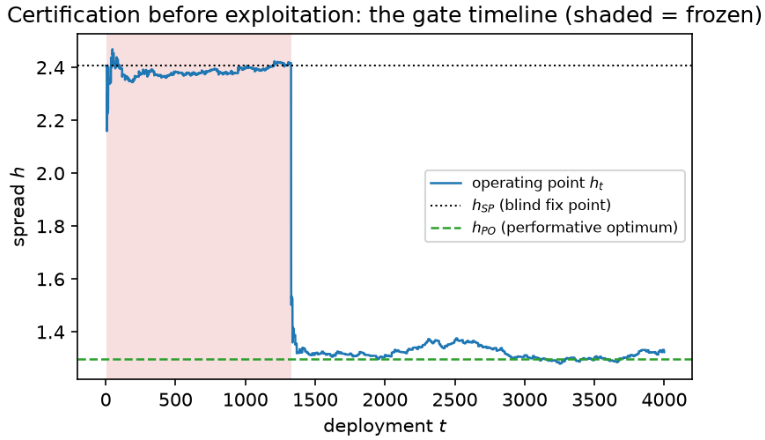

Figure 5: Certification before exploitation (L4, P7.1). Operating point h_t of the certified agent over 4,000 deployments. In the shaded interval the confidence sequence cannot yet certify that correcting is safe, so the agent stays frozen near the blind fixed point h_{SP}; once certified, it moves to the performative optimum h_{PO} and remains inside the trust region (Table 1: 0 violations).

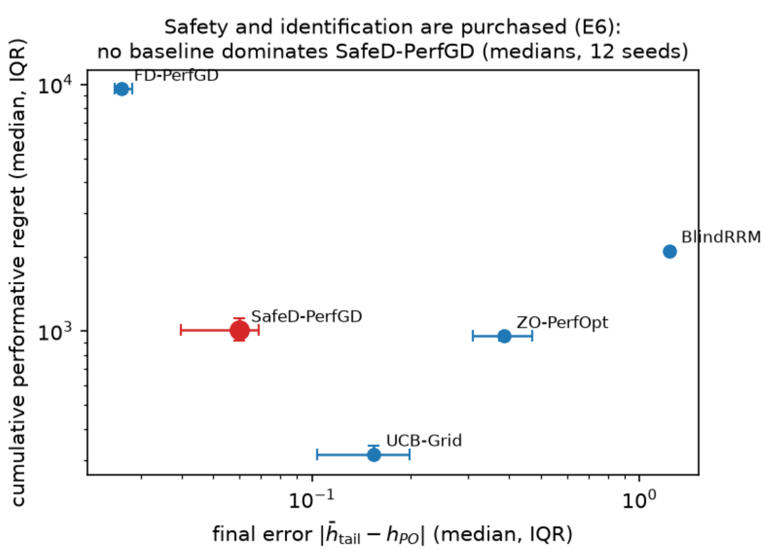

Figure 6: Baseline comparison (E6). Median final error $|\bar{h}_{\mathrm{tail}} - h_{\mathrm{PO}}|$ against median cumulative performative regret over 12 seeds (bars: interquartile range) for the certified agent (SafeD-PerfGD) and four baselines. No baseline dominates it: FD-PerfGD attains lower error at roughly ten times the regret, and UCB-Grid lower regret at higher error. Safety and identification are purchased, not free.

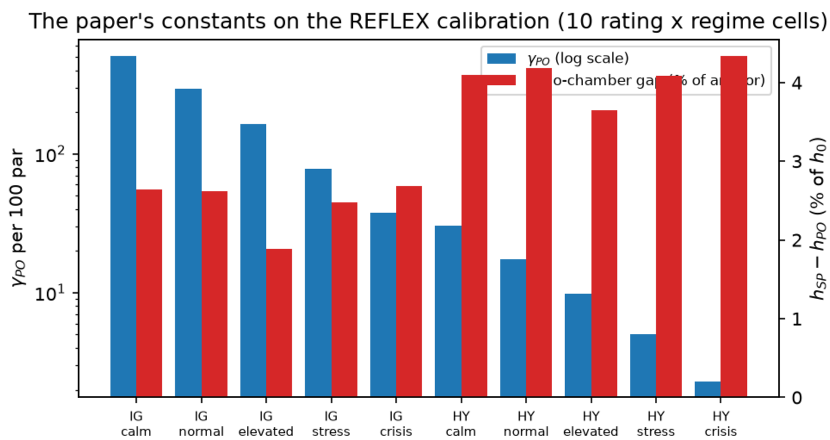

Figure 7: Real-data calibration (Table 1, calibration row). The curvature γ_{PO} (log scale, left axis) and the echo-chamber gap $h_{\mathrm{SP}} - h_{\mathrm{PO}}$ as a percentage of the anchor (right axis) across the ten rating-by-regime cells of the REFLEX corporate-bond calibration [Nagarajan and Ashok, 2026] (IG: investment grade; HY: high yield). All ten cells reproduce within 0.8 percent.

F Reproducibility

What ships. The supplementary repository contains the derivation documents and the theorem register (26 results, 38 numerical derivation checks), the simulation pipeline (environments, estimators, designs, agents, baselines; 36 measured-versus-predicted rows; 9 unit tests), the real-data port with its 10-cell validation, and the eleven-pivot record in the pipeline README. Both verification suites run in continuous integration on every push in the project repository; the two workflow files are not part of this anonymized copy. The structural market used throughout, and the corporate-bond calibration behind the real-data row of Table 1, are those of REFLEX [Nagarajan and Ashok, 2026], on which this work builds. All results are deterministic from seeds, `numpy`-only, and CPU-only. Reported final errors are 300-step tail averages over the deployed path, stable within 0.01 under horizon perturbation, with a worst measured spread of 0.0062.

What is stated here but not proved here. This version proves Theorems 1 and 2, the two results the body argues from. The minimax floors and the design and certification results (T3, T4, T5a, C5.1, T6, T7, L4, P7.1) are proved in the supplementary repository, in `mlxor-derivations/latex/proofs.tex` and the corresponding derivation documents, with the register recording each result's proof status.

Provenance of the numbers. Every quantity in the body and in Table 1 is copied from a committed artifact: `results/RESULTS.md` for the harness rows, `results/OPEN1.md` for the reach, the witness, and the search history, `results/REALDATA.md` for the calibration port, `results/STABILITY.md` for the tail-average stability, `results/full_run.log` for quantities not surfaced in the summary tables, and `THEOREMS.md` for the register. The runs behind them were executed and recorded on 1 September 2026. This version re-narrates them and does not re-run them, since none of its changes touch their inputs.